Time Series Representation: Transformations, Windows, and Event Detection

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Time Series Processing

Transform the raw series into an analyzable object, ready for prediction and event detection.

Before modeling, the series undergoes a transformation $\mathcal{T}$ that converts it into a new representation:

$$Y_t = \mathcal{T}(X_t)$$

Where X_t is the observed value at time t, Y_t is the transformed representation, and $\mathcal{T}$ is the chosen transformation. The choice of $\mathcal{T}$ determines which patterns and deviations become visible — and, therefore, what can be predicted or detected.

> This lecture is the **operational bridge** between understanding a series and using it for prediction and event detection. Transformations are the mechanism that converts structural knowledge of the series into actionable representations for modeling.

Temporal Aggregation

- Groups observations at a lower frequency.
- Reduces noise and simplifies analysis.
- May also remove important details for short-duration events.

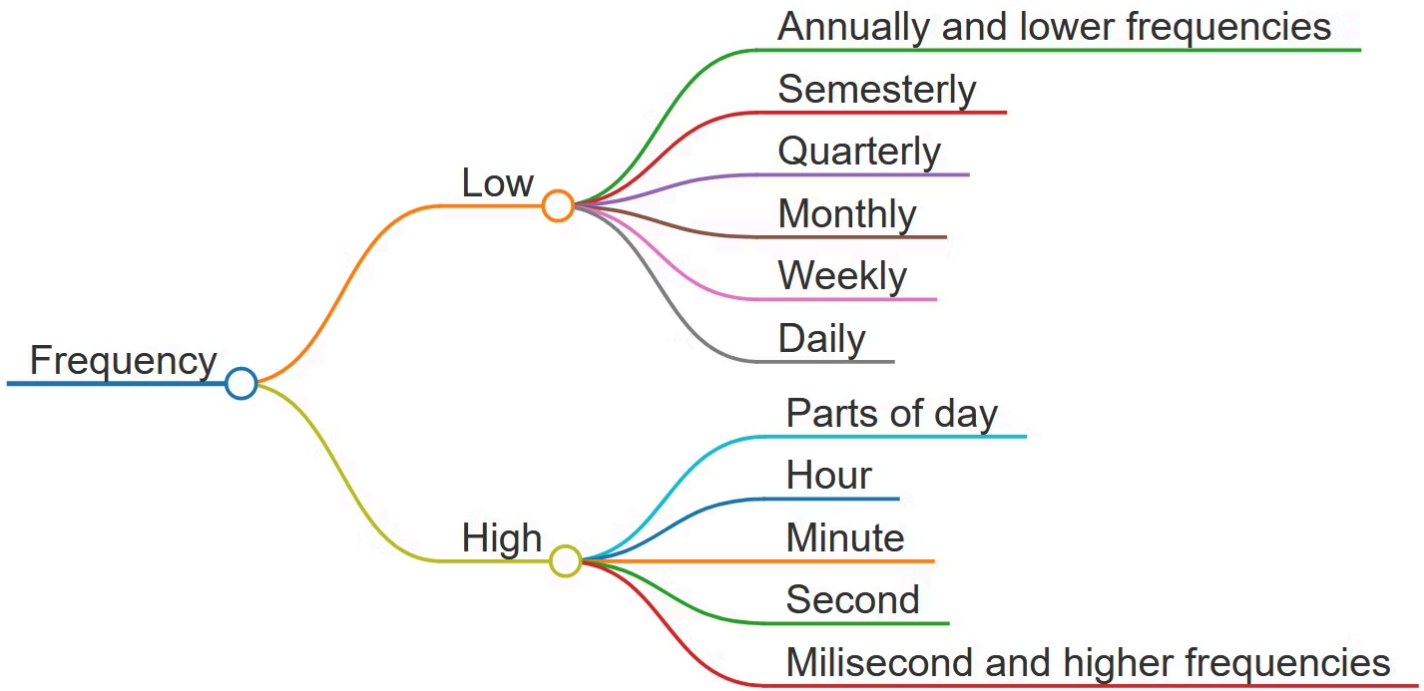
Formally, given an aggregation factor m:

$$Y_k = \mathcal{A}(X_{(k-1)m+1}, \ldots, X_{km})$$

where m is the aggregation factor, k is the index of the aggregated period, and $\mathcal{A}$ is the aggregation function (e.g., mean, sum, maximum).

For mean aggregation:

$$Y_k = \frac{1}{m} \sum_{j=1}^{m} X_{(k-1)m+j}$$



Temporal aggregation changes the frequency level used in the analysis.

Temporal Pre-processing

Operational transformations that adjust the **scale** and **structure** of the series before analysis — changing how deviations and events appear in the data, without altering their temporal sequence.

Normalization

$$X_t^{(norm)} = \frac{X_t - \mu}{\sigma}$$

Where μ is the series mean and σ is the standard deviation.

Requires $\sigma > 0$; if $\sigma = 0$, z-score normalization is undefined.

Differencing

$$\Delta X_t = X_t - X_{t-1}$$

Change between consecutive time steps.

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INPUT: X (time series of length T)
PRE-CONDITION: observed series; missing values handled or
               removed before numerical transformations

STEP 1 — Normalization
  μ ← mean(X)
  σ ← std_deviation(X)
  IF σ = 0: halt — normalization undefined
  FOR t from 1 to T:
    X_t^(norm) ← (X_t - μ) / σ

STEP 2 — Differencing
  FOR t from 2 to T:
    ΔX_t ← X_t - X_{t-1}

OUTPUT: X^(norm) (normalized series), ΔX (differenced series)
POST-CONDITION: X^(norm) has mean ≈ 0 and σ ≈ 1;
                ΔX removes linear trends
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Temporal Engineering and Deviation Modeling

Models generate a representation of the series, and deviations are identified relative to the model. The choice of representation is decisive.

Residual Calculation

$$R_t = X_t - \hat{X}_t$$

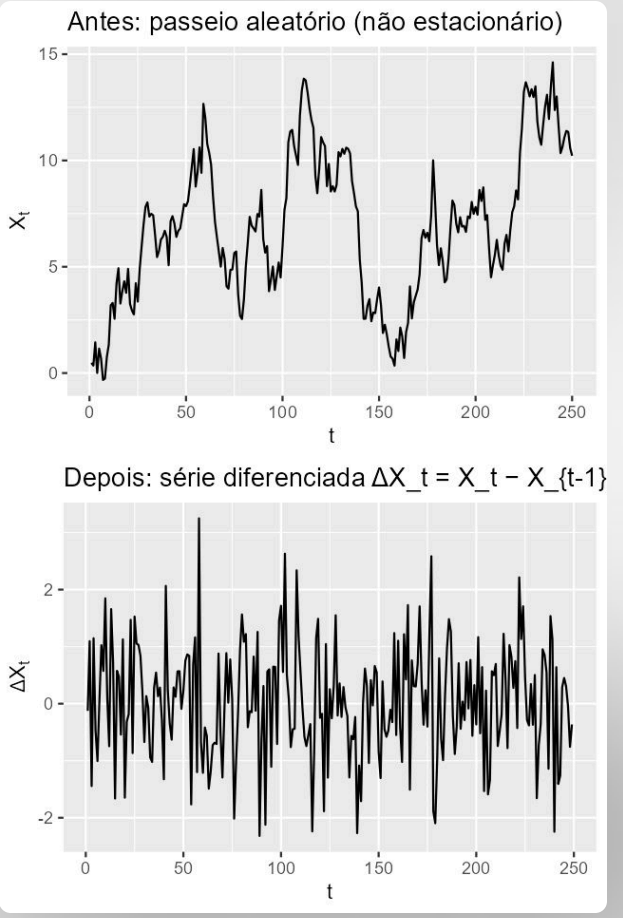
Where X_t is the observed value and $\hat{X}_t$ is the value estimated by the model.

Deviation Identification

$$|R_t| > h$$

A deviation occurs when the residual R_t exceeds the threshold h.

We first define the expected behavior of the series through a model. We then measure how far the actual data deviates from that expected behavior.



Differencing as a Transformation for Stationarity

Non-stationary series have a mean or variance that changes over time, making them unsuitable for direct modeling. Differencing is the most common transformation used to stabilize a series: it subtracts each observation from the previous one, removing systematic trends and making the series more amenable to modeling.

$$Y_t = X_t - X_{t-1}$$

Differencing is a preparation tool — it transforms the series so that the fundamentals of stationarity apply. The operational details of when and how to difference, including the choice of differencing order, will be explored in depth in upcoming presentations.

Temporal Transformations

Instead of working with levels, many analyses operate on **changes**: simple and logarithmic returns highlight relative variations between consecutive periods.

Simple Return

$$R_t = \frac{X_t - X_{t-1}}{X_{t-1}}$$

Measures the percentage change between consecutive periods. Requires $X_{t-1} \neq 0$.

Logarithmic Return

$$r_t = \log(X_t) - \log(X_{t-1})$$

More convenient mathematical properties for analysis. Requires $X_t > 0$ for all t.

Fourier Transform

Every time series can be viewed from two angles: in the time domain, we observe values across instants; in the frequency domain, we see which cycles compose the series — and with what intensity.

The discrete Fourier transform performs this change of perspective:

$$C_k = \sum_t X_t \, e^{-i2\pi kt/n}$$

Each coefficient C_k captures the energy associated with frequency k, revealing periodic patterns invisible in the raw series.

Fourier shows which frequencies exist in the series, but does not preserve **when** each oscillation occurs — temporal localization is lost in the transformation.

Time Domain

Shows **when** something happens. Useful for detecting point changes and trends.

Frequency Domain

Shows **how** the series oscillates. Useful for isolating regular cycles and separating signal from noise.

Each representation reveals a different type of pattern. The loss of temporal localization in Fourier — not knowing **when** each frequency occurs — motivates the use of wavelets, which combine temporal and spectral resolution.

Wavelet Decomposition

Wavelets offer an alternative representation of the series that combines time and frequency simultaneously. Unlike Fourier, which dissolves temporal information, wavelets make localized deviations visible — anomalies that occur at specific moments and at specific scales.

$$W(a,b) = \int X(t)\,\psi_{a,b}(t)\,dt$$

The parameter a controls the scale (wavelet width) and b controls the temporal position.

Wavelets simultaneously preserve time and scale information — something Fourier cannot achieve. The choice of representation determines what is detectable.

Temporal Localization

Reveals *when* a deviation occurs — something that the Fourier representation does not preserve

Multi-scale Analysis

Reveals *at what scale* the deviation manifests — distinguishing rapid fluctuations from slow changes

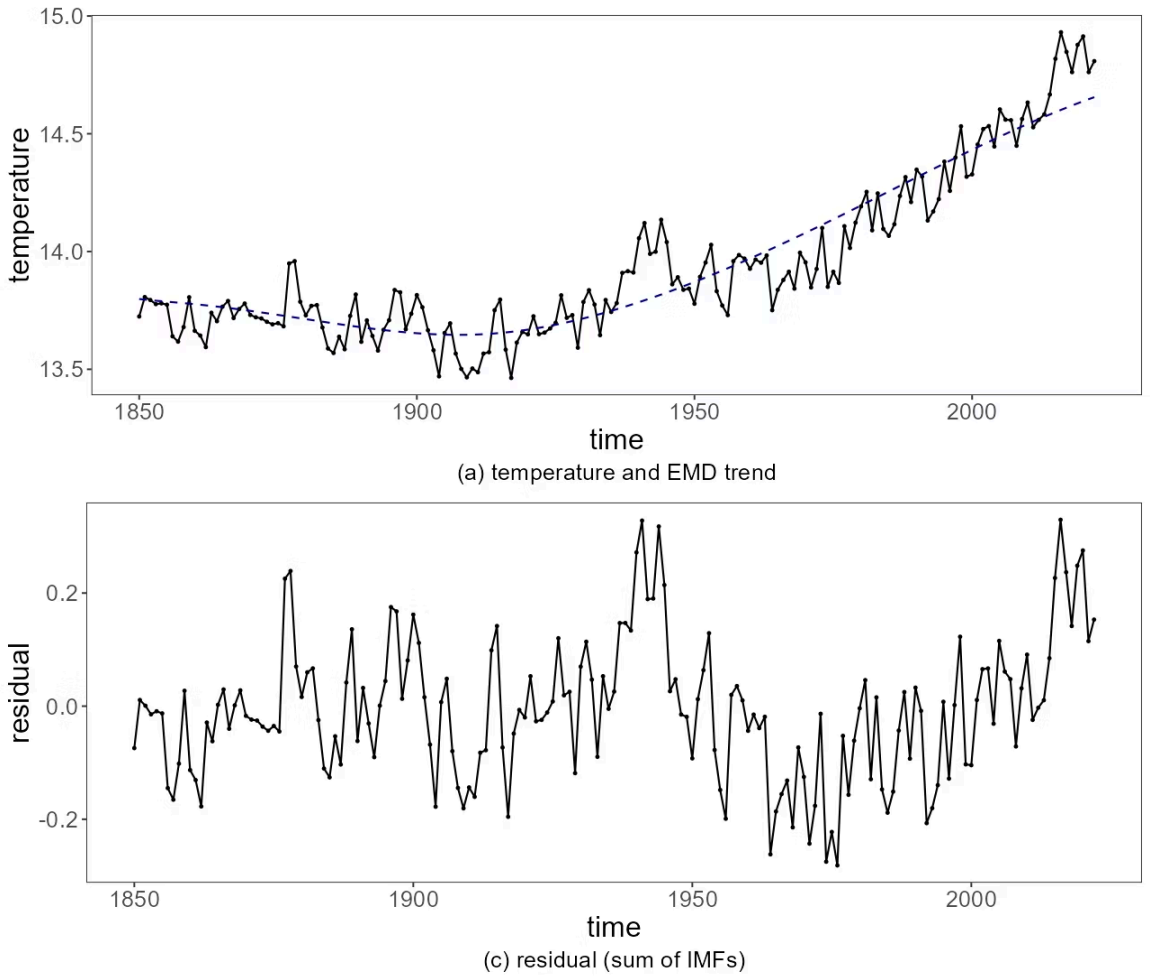
EMD Decomposition

EMD decomposes a signal into oscillatory components called IMFs (Intrinsic Mode Functions):

$$X(t) = \sum_{i=1}^{n} \mathrm{IMF}_i(t) + r(t)$$

where $\mathrm{IMF}_i(t)$ are the intrinsic mode functions and $r(t)$ is the residual (long-term trend).

Each IMF represents a distinct oscillation scale. The residual $r(t)$ captures the global trend of the series.



EMD applied to YGT: residual trend component and first IMF.

Local Representation of the Series

Dividing the series into subsequences reveals **local patterns** invisible from a global perspective. The subsequence $X_{t:t+w-1}$ defines a segment of length w starting at t:

$$X_{t:t+w-1} = (X_t, X_{t+1}, \ldots, X_{t+w-1})$$

Each subsequence transforms a portion of the series into a comparable object: it is possible to measure similarity, distance, local shape, or local deviation between windows. This allows detecting behavioral changes confined to specific intervals — and is the basis of the sliding window mechanism.

Sliding Windows

The window size w defines **which scale of behavior becomes visible** — each choice induces a different local representation of the series.

$$\mathcal{W}_w(X_t) = \{X_{t:t+w-1}\} \text{ for each position } t$$

Short Windows

Point anomalies and abrupt changes

Medium Windows

Transient patterns and regime shifts

Long Windows

Large-scale trends and cycles

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INPUT: X (series of length T), w (window size), h (threshold)
PRE-CONDITION: w ≥ 2; T > w; online detection scenario (one-step-ahead)

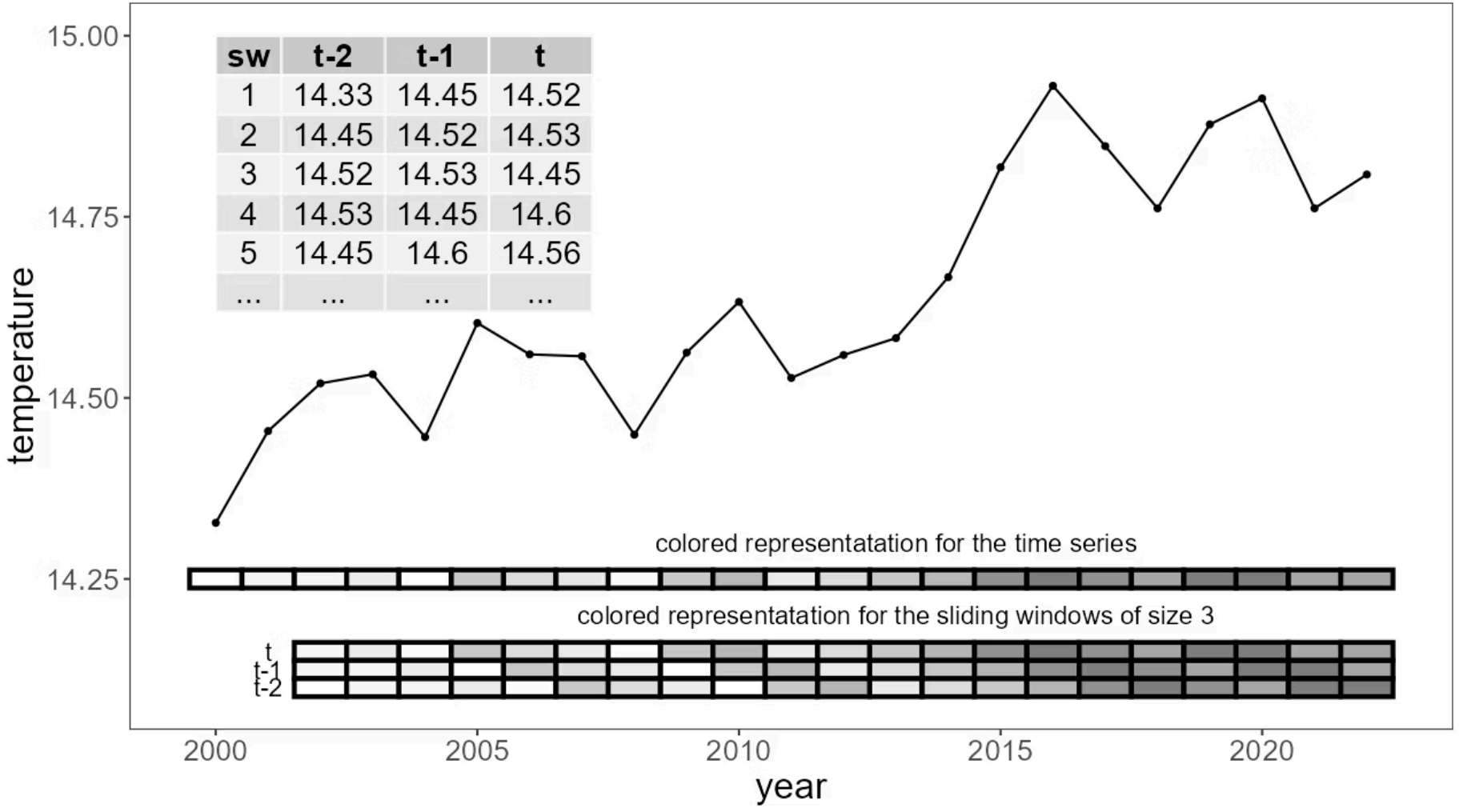
STEP 1 — Past window sliding
  FOR t from w+1 to T:
    W_t ← X[t-w : t-1]   // window of w points preceding t

STEP 2 — Past window statistic computation
  FOR each t:
    μ_t ← mean(W_t)
    σ_t ← std_deviation(W_t)
    IF σ_t = 0: σ_t ← 1   // protection against division by zero

STEP 3 — Current point evaluation
  FOR each t:
    IF |X_t - μ_t| > h · σ_t:
      mark t as deviation   // X_t does not belong to the reference window

OUTPUT: list of time instants t marked as deviation
POST-CONDITION: deviations detected without temporal leakage; X_t evaluated
using only information prior to t
```

Subsequences and sliding windows



A series can be decomposed into windows that shift along the time axis.

Temporal Workflow

The processing of time series follows a **pipeline of transformations**. The steps are typically sequential, but may involve choices, alternatives, and feedback loops — each decision changes the type of pattern or deviation that becomes visible.

Raw Series

X_t — original signal, with noise and arbitrary scale

Transformation

$X_t^{(tr)}$ — normalization, differencing, or decomposition

Representation

$X_t^{(rep)}$ — sliding window or features extracted from the series

Model

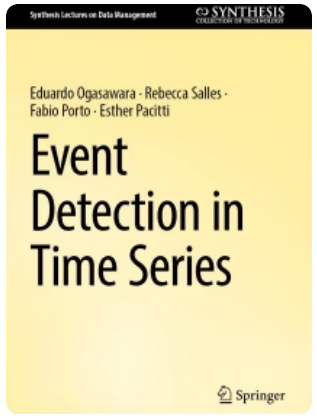
$\hat{X}_t$ — fitting or forecasting over the representation

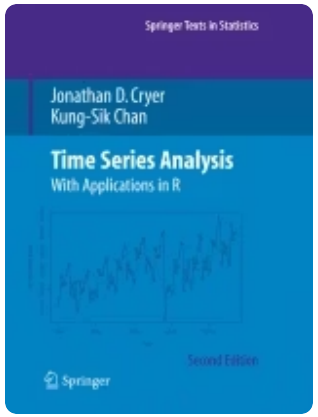
Analysis

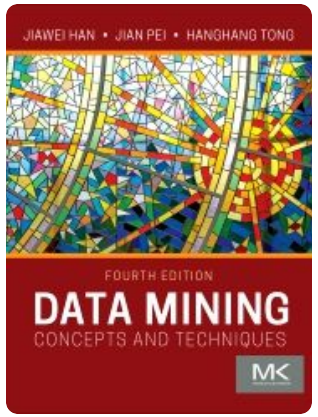
$R_t = X_t - \hat{X}_t$ (residual); $E_t = \mathbf{1}\{|R_t| > h\}$ (deviation indicator)

The choice of pipeline determines what is detectable. This lecture provides the tools to build that pipeline — from the initial transformation to the generation of forecasts and the detection of events. Each step is a decision about what matters to observe.

Bibliographic References







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